

Ji Jade Qiu

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RESEARCH INTERESTS

LLM-based systems and reliable evaluation; generalization theory of neural networks; multi-agent LLM communication (current interest)

EDUCATION

Columbia University

Aug 2025 - Dec 2026

Master of Science in Applied Analytics

GPA: 3.70

Relevant Coursework: Applied Machine Learning II (A+), Applied Data Science (A), Fundamentals of Data Engineering (A-)

Xi'an Jiaotong-Liverpool University

Aug 2021 - May 2025

Bachelor of Science in Financial Mathematics (honors)

SKILLS

Programming: Python, SQL

ML / DL: PyTorch, PyTorch Geometric, scikit-learn, XGBoost

LLMs: Gemini API (schema-constrained structured extraction)

Infrastructure: HPC cluster (SLURM), Git, SQLite, AWS S3

PUBLICATIONS

Learning on Graphs Conference (LoG) 2026

Accepted (Poster)

Proceedings Track

Generalization of Hypergraph Neural Networks via VC Dimension and Rademacher Complexity

Yamini Chitika, [Ji Qiu](#), Chunjiang Zhu

Characterized the generalization ability of hypergraph neural networks (HNNs) by tying their VC dimension to a two-stage hypergraph Weisfeiler–Leman (HWL) coloring: cast HNNs exactly as feedforward networks to obtain bounds logarithmic in HWL color complexity and polynomial in parameter count, showed that VC dimension is tightly controlled by weight bitlength when hypergraph order is unbounded, and gave matching upper and lower bounds via the number of HWL-distinguishable hypergraphs when it is bounded. Complemented these worst-case results with an HWL color-class Rademacher bound that is tight up to a constant for balanced color classes, and confirmed the predicted trends on ten real and synthetic hypergraph datasets through width, depth, and shattering experiments.

Contribution: designed and ran the full empirical study (width/depth sweeps and shattering experiments) on the Wahab HPC cluster.

RESEARCH EXPERIENCE

Department of Computer Science, Old Dominion University

May 2026 – Aug 2026

Summer Research Intern

- **Automated LLM-based Literature Mining for a Large-Scale Hepatotoxicity Dataset**
(advised by Prof. Chunjiang Zhu)

Built an end-to-end pipeline for automatically mining hepatotoxicity findings from the biomedical literature, spanning literature retrieval, structured extraction, and downstream validation. Collected

53.6k papers from PubMed and PMC with resumable, rate-limit-aware downloads and daily incremental updates, then used Gemini 2.5 with a controlled 14-field schema and anti-hallucination constraints to extract 189k structured findings across 17k chemicals. Validated the resulting dataset through per-marker QSAR modeling, comparing Graphormer, GraphGPS, and GraphSAGE against RandomForest and XGBoost, with manual review confirming 95.4% core-triple accuracy. First-authored a paper submitted to WSDM 2027.

Department of Financial Mathematics, XJTU

Dec 2024 – Apr 2025

Final Year Project

- **LSTM-based Equity Portfolio Optimization with Shannon Entropy Regularization**
(advised by Prof. Xiangqian Sun)

Developed a deep learning framework for large-cap equity portfolio optimization by combining 1) LSTM-based return forecasting with 2) Shannon entropy regularization and an Auto-regressive Moving Pointer Model (AMPM). The framework uses entropy-based diversification constraints to reduce concentration risk while preserving return potential, linking predictive modeling directly to portfolio construction. Built rolling-window validation pipelines for robust out-of-sample backtesting, achieving 10% annualized returns with a Sharpe ratio of 0.9 and a maximum drawdown 15% lower than the market-cap-weighted benchmark.

Pacific Insurance Company

May 2024 – Aug 2024

Economic Scenario Generator Project

- **Economic Scenario Generator for Insurance Risk Management**

Developed a Python-based economic scenario generator to simulate stochastic paths of key market variables for insurance asset–liability and solvency risk analysis. Calibrated a nonlinear state-space model with an Unscented Kalman Filter (UKF), producing stable parameter estimates in regimes where linear Kalman and EKF approaches degrade. Fitted a Cox–Ingersoll–Ross (CIR) short-rate model to historical interest-rate data and modeled equity dynamics using geometric Brownian motion (GBM). Evaluated the interest-rate model on a held-out test set, achieving a short-rate RMSE of 15 basis points.

COMMUNITY SERVICE

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| • Outstanding Class Leader | 2021 - 2023 |
| • Student Representative, Department of Financial Mathematics | 2022 - 2023 |
| • President, Student Career Development Association, Information Division | 2021 - 2022 |